

However, the idea of maximizing or minimizing something is extremely common. Businesses attempt to choose prices to maximize profits. Manufacturers attempt to choose practices that maximize efficiency and minimize defects. Engineers attempt to choose design features that maximize performance or minimize drag or cost. Economics is largely structured around maximization and minimization problems: maximizing utility especially, and also maximizing dollar amounts like GDP and revenue, and minimizing estimation error. Machine learning and statistics rely on minimization for the bulk of their methods; they minimize a “loss function” or an error metric. For each of these, there is the potential to use a hill-climbing solution like gradient ascent or descent to get to an optimal solution.

Even in everyday life, we choose how much money to spend to maximize achievement of our financial goals. We strive to maximize happiness and joy and peace and love and minimize pain and discomfort and sadness.

For a vivid and relatable example, think of being at a buffet and seeking, as all of us do, to eat the right amount to maximize satisfaction. If you eat too little, you will walk out hungry and you may feel that by paying the full buffet price for only a little food, you haven’t gotten your money’s worth. If you eat too much, you will feel uncomfortable and maybe even sick, and maybe you will violate your self-imposed diet. There is a sweet spot, like the peak of the tax/revenue curve, that is the exact amount of buffet consumption that maximizes satisfaction.

We humans can feel and interpret sensory input from our stomachs that tells us whether we’re hungry or full, and this is something like a physical equivalent of taking a gradient of a curve. If we’re too hungry, we take some step with a predecided size, like one bite, toward reaching the sweet spot of satisfaction. If we’re too full, we stop eating; we can’t “un-eat” something we have already eaten. If our step size is small enough, we can be confident that we will not overstep the sweet spot by much. The process we go through when we are deciding how much to eat at a buffet is an iterative process involving repeated direction checks and small steps in adjustable directions—in other words, it’s essentially the same as the gradient ascent algorithm we studied in this chapter.